PROJECT 2 OF 2 · COMPUTER VISION + 3D RECONSTRUCTION

Roadside Tree & Power-Line Risk Map

Which roadside trees could fall, or are growing into the wires?

AI turns phone video, filmed from a moving vehicle, into a map of roadside trees, each scored for **fall risk** and for branches too close to **power lines**.

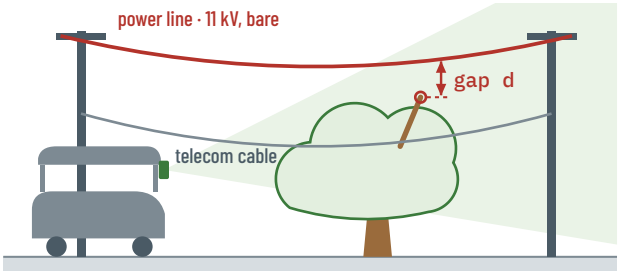
ROUTES	About 100 km: urban (near UoM), suburban, hill-country
USERS	Councils, RDA, CEB, LECO
KIT	Phones, damped mount, rangefinder
PLAN	Oct 2026 – Jun 2027 · 3 members

THE PROBLEM

No low-cost way exists to survey every roadside tree and nearby wires, so risky trees are found late or felled in bulk.

- 4 Jun 2026 — A eucalyptus fell on a moving three-wheeler at Nawalapitiya; three killed.
- 30 May 2025 — About 50,000 power breakdowns in 24 hours of monsoon rain and wind; fallen trees blocked Colombo roads.
- 2023–24 — Colombo council flags 558 dangerous century-old trees; in 2024, storms mostly felled trees not flagged as risky.
- Oct 2023 — A State Minister orders removal of roadside trees at risk of falling.

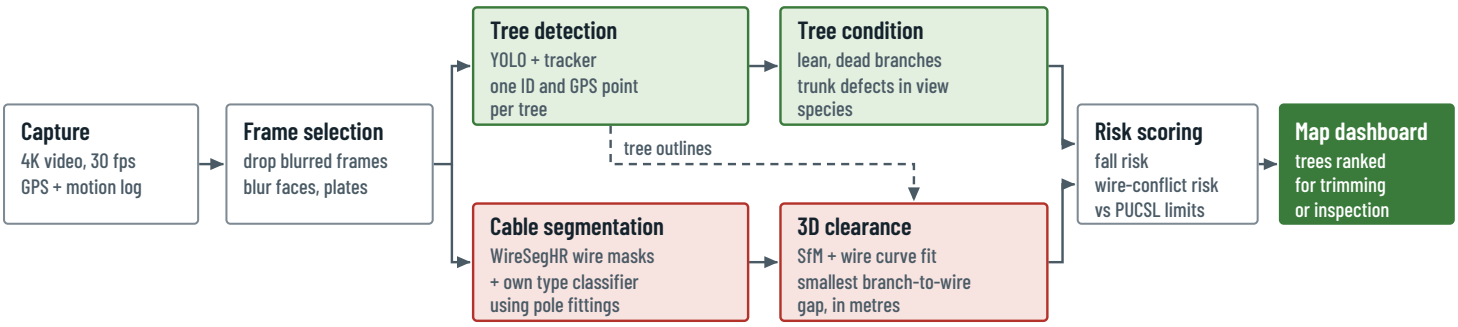
THE KEY IDEA



The moving phone sees each tree from many angles: COLMAP rebuilds the tree, each wire is triangulated across frames, and GPS plus known sizes such as poles fix the scale of *d*.

HOW IT WORKS

eight stages, a tree lane and a cable lane



Clearance needs both lanes; results are checked against laser gaps, expert risk ratings and LECO or CEB cable labels.

WHAT IS NEW

● covered ○ not

We found no study measuring branch–wire gaps from street-level phone or dashcam video, or in Sri Lanka.

STUDY	STREET LEVEL	GAP IN METRES	PHONE OR DASHCAM	FALL RISK	SRI LANKA
Oliveira et al. 2022 yes/no	●	○	○	○	○
Silva et al. 2026 height only	●	○	○	○	○
Rong et al. 2020 on towers	○	●	○	○	○
Bahreini et al. 2024 mobile LiDAR	●	●	○	○	○
Joshi et al. 2025 dashcam, no wires	●	○	●	○	○
Guruge & Herath 2026 Colombo, GIS	●	○	○	●	●
This project	●	●	●	●	●

HOW WE PROVE IT

on held-out routes

TESTED	METRIC	AGAINST
Trees	Precision, recall, mAP	Public-data-only model
Cables	IoU, F1, cable-type accuracy	WireSegHR-only model
Clearance	Error in m vs two-point laser	Image-overlap check
Fall risk	Kappa vs expert ratings	Rule-based checklist
Whole system	Survey time per km	Manual survey

RESEARCH QUESTIONS AND OWNERS

- A+B **RQ1** How accurately can AI find trees and overhead cables in phone video, and tell power cables from telephone and fibre?
- B **RQ2** How accurately can the branch-to-wire gap be estimated from video, against on-site measurements?
- A **RQ3** Can signs visible from the road (lean, dead branches, trunk defects, species) predict an expert's risk rating?
- C **RQ4** How much accuracy do models trained abroad lose on Sri Lankan roads, and how much local data closes the gap?

A trees and fall risk **B** cables and clearance
C data, privacy, dashboard

LEGAL LIMITS FOR THE GAP

LINE · PUCSL GUIDELINE V4.0, 2020	VERTICAL	HORIZONTAL
Low voltage, bare	2.7 m	1.5 m
Low voltage, insulated	0.15 m	0.15 m
11 kV	2.7 m	1.5 m
33 kV	3.7 m	2.9 m

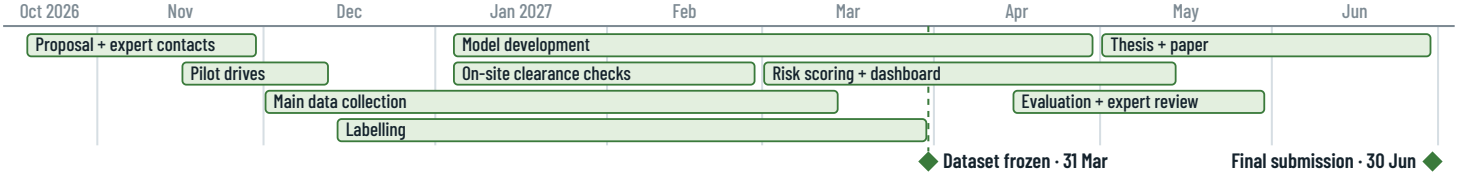
Bare and insulated limits differ up to 18×: a new sub-task.

MAIN RISKS

- ▲ **Live wires.** A passenger films; gaps measured from the ground.
- ▲ **Privacy.** Faces and plates blurred; ethics review first.
- ▲ **Wrong gaps.** Screening aid only; crews confirm first.

PLAN

proposed; dates assumed, adjust to faculty deadlines



SOURCES PUCSL 2020 · Daily Mirror Oct 2023, Dec 2023, Jun 2024, Jun 2026 · Daily FT 2025 · Oliveira et al. 2022 · Silva et al. 2026 · Rong et al. 2020 · Bahreini et al. 2024 · Joshi et al. 2025 · Guruge & Herath 2026 · WireSegHR

6.80°N 79.90°E · Moratuwa routes